

A CROSS-ENTROPY NOISE PROPAGATION OPERATOR AND THE EFFECTIVE RANK AT THE NEURAL COLLAPSE FIXED POINT

LIGHTMAN CHANG

ABSTRACT. We extend the noise-propagation framework of [1]—originally formulated for the squared loss in regression—to multi-class classification under the cross-entropy loss. The technical step is to derive a Fisher-corrected noise propagation operator $\mathbf{M}_{\text{CE}}$ directly from the cross-entropy Hessian $\nabla_{\theta}^2 \ell_{\text{CE}} = \mathbf{J}^\top \mathbf{H} \mathbf{J}$, where $\mathbf{H}(\mathbf{x}) = \text{diag}(\mathbf{p}(\mathbf{x})) - \mathbf{p}(\mathbf{x})\mathbf{p}(\mathbf{x})^\top$ is the output-space Fisher matrix at the softmax probabilities $\mathbf{p}(\mathbf{x})$. Our main result (Theorem 4.1) is that $\text{tr}(\mathbf{M}_{\text{CE}}) < \infty$ if and only if the underlying power-law exponents satisfy $\beta > \alpha + 1/2$, identical to the trichotomy for MSE. We then study the limit of $\mathbf{M}_{\text{CE}}$ along a sequence of trained predictors approaching the Neural Collapse fixed point of Papanicolaou–Han–Donoho. Although $\mathbf{H}$ becomes singular in the strict one-hot limit, the limiting operator that controls the test-risk effective rank is well-defined on the $(K - 1)$ -dimensional zero-sum subspace fixed by the simplex equiangular tight frame, and Theorem 5.3 computes its effective rank exactly: it equals $K - 1$, the number of classes minus one. The argument is purely linear-algebraic given the simplex ETF geometry, and yields a dimension-free generalization guarantee for K -class classification at the terminal phase of training.

1. INTRODUCTION

The unified bound of [1] on the test risk of overparameterized regression involves a noise-propagation operator $\mathbf{M} = \Sigma^{-1} \Gamma \Sigma^{-1}$ built from the SVD of the training Jacobian $\mathbf{J}$ and the test–train cross-correlation Γ of gradient features. The trichotomy $\text{tr}(\mathbf{M}) < \infty \iff \beta > \alpha + 1/2$, under power-law decay $\sigma_j \asymp j^{-\alpha}$ and $\varrho_j := \sqrt{\Gamma_{jj}} \asymp j^{-\beta}$, characterizes benign, tempered, and catastrophic regimes for squared loss.

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The same generalization story should apply to classification, but the squared-loss derivation does not directly transfer: cross-entropy introduces a nonlinear softmax link, and its parameter Hessian is $\mathbf{J}^\top \mathbf{H} \mathbf{J}$ rather than $\mathbf{J}^\top \mathbf{J}$. The natural question is whether a Fisher-corrected operator preserves the trichotomy and admits an exact effective-rank calculation at the Neural Collapse (NC) fixed point of [8].

This paper answers both questions affirmatively. We define $\mathbf{M}_{\text{CE}}$ as the natural generalization of $\mathbf{M}$ obtained by replacing $\mathbf{J}^\top \mathbf{J}$ with the cross-entropy Hessian (see Definition 3.1 and the derivation that precedes it). We then show:

- (i) The CE trichotomy $\text{tr}(\mathbf{M}_{\text{CE}}) < \infty \iff \beta > \alpha + 1/2$ holds whenever the predicted probabilities are bounded away from the one-hot vertices uniformly along the relevant directions (Theorem 4.1).
- (ii) Approaching the NC simplex equiangular tight frame fixed point, the limiting operator on the zero-sum invariant subspace has effective rank exactly $K - 1$ (Theorem 5.3), where K is the number of classes.

The first result extends the trichotomy across loss functions; the second states that, at the terminal phase of training, the generalization-controlling rank is set by the number of classes minus one, regardless of the input dimension. We work conditional on the presence of NC and characterize what this implies for the effective rank, rather than proving NC.

Relation to prior work. The Neural Collapse phenomenon was identified empirically in [8] and characterized mathematically in the unconstrained-features setting by [6, 5, 7]. The cross-entropy Fisher matrix $\mathbf{H} = \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$ and its degeneracy at one-hot vertices is classical (see [3]); we use only the finite-rank facts about $\mathbf{H}$ and the simplex ETF geometry.

Organization. Section 2 fixes notation and recalls the classification setting. Section 3 derives $\mathbf{M}_{\text{CE}}$ from the CE Hessian. Section 4 proves the CE trichotomy. Section 5 derives the effective-rank formula at the NC fixed point. Section 6 discusses limitations.

2. SETTING AND NOTATION

Let $K \geq 2$ be the number of classes. The predictor has K -dimensional output $f(\mathbf{x}; \theta) = (f_1(\mathbf{x}; \theta), \dots, f_K(\mathbf{x}; \theta)) \in \mathbb{R}^K$, softmax probabilities $p_k(\mathbf{x}; \theta) = e^{f_k} / \sum_{k'} e^{f_{k'}}$, and one-hot labels $\mathbf{y}_i \in \{e_1, \dots, e_K\}$, where e_k is the k -th standard basis vector. Training data $S = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$ are

i.i.d. The cross-entropy loss is

$$\ell_{\text{CE}}(f, \mathbf{y}) = - \sum_{k=1}^K y_k \log p_k(f).$$

A direct computation gives the well-known output-space Hessian

$$(1) \quad \nabla_f^2 \ell_{\text{CE}}(f, \mathbf{y}) = \mathbf{H}(\mathbf{x}) := \text{diag}(\mathbf{p}(\mathbf{x})) - \mathbf{p}(\mathbf{x})\mathbf{p}(\mathbf{x})^\top,$$

which is symmetric positive semidefinite, has rank $K - 1$ for any $\mathbf{p}$ in the open simplex (it has a single zero eigenvalue along $\mathbf{1}$), and rank 0 at every one-hot vertex $\mathbf{p} = e_k$ (since then $\mathbf{H} = e_k e_k^\top - e_k e_k^\top$, adjusted gives the degenerate form $\text{diag}(e_k) - e_k e_k^\top$, which vanishes on the support of e_k). The Fisher matrix $\mathbf{H}(\mathbf{x})$ is the building block of the parameter Hessian:

$$(2) \quad \nabla_\theta^2 \ell_{\text{CE}}(\mathbf{x}, \mathbf{y}; \theta) = \mathbf{J}(\mathbf{x})^\top \mathbf{H}(\mathbf{x}) \mathbf{J}(\mathbf{x}) + (\text{label-dependent rank-1 term}),$$

where $\mathbf{J}(\mathbf{x}) \in \mathbb{R}^{K \times p}$ is the per-sample Jacobian of $f(\mathbf{x}; \cdot)$ with respect to θ . The label-dependent term vanishes in expectation over labels at any predictor that is exact on the population (or at any limit point with vanishing gradient), and we drop it from this point onward.

We adopt the notation of [1]. Let $\mathbf{J}_{\text{CE}} \in \mathbb{R}^{nK \times p}$ denote the stacked training Jacobian with rows $\nabla_\theta f_k(\mathbf{x}_i; \theta^*)^\top$ ranging over $(i, k) \in [n] \times [K]$, and let $\mathbf{H} = \bigoplus_{i=1}^n \mathbf{H}(\mathbf{x}_i)$ denote the block-diagonal sample Fisher. Denote the cross-class test–train cross-correlation

$$(3) \quad (\mathbf{\Gamma}_{\text{CE}})_{(j,k),(j',k')} = \mathbb{E}_{\mathbf{x}}[\psi_{j,k}(\mathbf{x})\psi_{j',k'}(\mathbf{x})], \quad \psi_{j,k}(\mathbf{x}) = \nabla_\theta f_k(\mathbf{x}; \theta^*)^\top \mathbf{v}_j,$$

where $\{\mathbf{v}_j\}$ is the right SVD basis of an appropriate factor of $\mathbf{J}_{\text{CE}}$ (see Section 3).

3. DERIVING $\mathbf{M}_{\text{CE}}$ FROM THE CROSS-ENTROPY HESSIAN

In MSE, the operator $\mathbf{M} = \mathbf{\Sigma}^{-1} \mathbf{\Gamma} \mathbf{\Sigma}^{-1}$ arises because the test risk decomposes as a quadratic form in the predicted residuals, weighted by the inverse of $\mathbf{J}^\top \mathbf{J}$ at the interpolant. Concretely, $\mathbf{\Sigma}$ comes from the SVD of the training Jacobian, and the noise-leakage variance equals $\sigma^2 \text{tr}(\mathbf{\Sigma}^{-1} \mathbf{\Gamma} \mathbf{\Sigma}^{-1})$.

For cross-entropy at a (near-)interpolating predictor, the linearized test risk is governed by the inverse of the parameter Hessian (2), that is $(\mathbf{J}_{\text{CE}}^\top \mathbf{H} \mathbf{J}_{\text{CE}})^{-1}$. Two natural factorizations arise:

- (1) Define the *Fisher-weighted Jacobian*

$$\tilde{\mathbf{J}} := \mathbf{H}^{1/2} \mathbf{J}_{\text{CE}} \in \mathbb{R}^{nK \times p},$$

- with SVD $\tilde{\mathbf{J}} = \tilde{\mathbf{U}} \tilde{\mathbf{\Sigma}} \tilde{\mathbf{V}}^\top$. Then $\mathbf{J}_{\text{CE}}^\top \mathbf{H} \mathbf{J}_{\text{CE}} = \tilde{\mathbf{V}} \tilde{\mathbf{\Sigma}}^2 \tilde{\mathbf{V}}^\top$, so the parameter Hessian's eigenstructure coincides with that of $\tilde{\mathbf{J}}^\top \tilde{\mathbf{J}}$.
- (2) The right singular vectors $\tilde{\mathbf{V}}$ provide the natural basis for gradient features $\tilde{\psi}_{j,k}(\mathbf{x}) := \nabla_{\theta} f_k(\mathbf{x}; \theta^*)^\top \tilde{\mathbf{v}}_j$, and the corresponding cross-correlation matrix is $\tilde{\mathbf{\Gamma}}_{(j,k),(j',k')} = \mathbb{E}_{\mathbf{x}}[\tilde{\psi}_{j,k}(\mathbf{x}) \tilde{\psi}_{j',k'}(\mathbf{x})]$.

These two ingredients give the operator that propagates label noise through the linearized CE risk:

Definition 3.1 (Cross-entropy noise propagation operator). The *cross-entropy noise propagation operator* is

$$(4) \quad \mathbf{M}_{\text{CE}} := \tilde{\mathbf{\Sigma}}^{-1} \tilde{\mathbf{\Gamma}} \tilde{\mathbf{\Sigma}}^{-1}, \quad \tilde{\mathbf{J}} = \mathbf{H}^{1/2} \mathbf{J}_{\text{CE}} = \tilde{\mathbf{U}} \tilde{\mathbf{\Sigma}} \tilde{\mathbf{V}}^\top,$$

where the inverse on the right is the Moore–Penrose pseudoinverse restricted to the range of $\tilde{\mathbf{\Sigma}}$.

Remark 3.2 (Equivalence with the form $\mathbf{F}^{-1} \mathbf{\Gamma} \mathbf{F}^{-1}$). Some prior write-ups state $\mathbf{M}_{\text{CE}} = \mathbf{F}^{-1} \mathbf{\Gamma}_{\text{CE}} \mathbf{F}^{-1}$ with $\mathbf{F} = \mathbf{H}^{1/2} \mathbf{\Sigma}_{\text{CE}}$. This is the same operator as (4) written in the basis of $\mathbf{J}_{\text{CE}}$ rather than $\tilde{\mathbf{J}}$: $\tilde{\mathbf{\Sigma}}$ are precisely the singular values of $\mathbf{H}^{1/2} \mathbf{J}_{\text{CE}}$, which equal the square roots of the eigenvalues of $\mathbf{J}_{\text{CE}}^\top \mathbf{H} \mathbf{J}_{\text{CE}}$. We retain Definition 3.1 as the canonical form because the factorization is unambiguous.

The justification for this definition is that, at the linearized order around a (near-)interpolant for CE, the noise-leakage variance in the test risk is $\sigma^2 \text{tr}(\mathbf{M}_{\text{CE}})$, with the same role for the Fisher matrix $\mathbf{H}$ as the role of $\mathbf{I}$ in MSE: $\mathbf{H}$ replaces $\mathbf{I}$ in the inner product on the output space.

4. CROSS-ENTROPY TRICHOTOMY

Theorem 4.1 (Cross-entropy trichotomy). *Suppose the singular values $\tilde{\sigma}_j$ of $\tilde{\mathbf{J}}$ and the diagonal $\tilde{\mathbf{\Gamma}}_{jj}$ exhibit power-law decay $\tilde{\sigma}_j \asymp j^{-\alpha}$, $\tilde{\varrho}_j := \sqrt{\tilde{\mathbf{\Gamma}}_{jj}} \asymp j^{-\beta}$, where the indexing combines (j, k) via the right singular basis of $\tilde{\mathbf{J}}$. Suppose furthermore that the predicted probabilities are uniformly bounded away from the one-hot vertices along the directions of decay, in the sense that there exists $h_{\min} > 0$ such that the diagonal of $\mathbf{H}$ along the right singular vectors of $\mathbf{J}_{\text{CE}}$ in the relevant basis is $\geq h_{\min}$ (uniformly over the indices that contribute to the sums below). Then*

$$\text{tr}(\mathbf{M}_{\text{CE}}) < \infty \iff \beta > \alpha + \frac{1}{2}.$$

Proof. By Definition 3.1, $\text{tr}(\mathbf{M}_{\text{CE}}) = \sum_j \tilde{\Gamma}_{jj} / \tilde{\sigma}_j^2 \asymp \sum_j j^{-2\beta} / j^{-2\alpha} = \sum_j j^{2(\alpha-\beta)}$, where we sum over all combined indices. The series converges iff $2(\alpha-\beta) < -1$, i.e. $\beta > \alpha + 1/2$. The $h_{\min}$ assumption ensures that the conversion between $\tilde{\sigma}_j$ and the singular values of $\mathbf{J}_{\text{CE}}$ inflates the constant by at most $h_{\min}^{-1/2}$ and does not affect the asymptotic exponent. $\square$

Remark 4.2 (On the uniform Fisher lower bound). The hypothesis $h_{\min} > 0$ is the cross-entropy analog of the “well-conditioned predictor” assumption: at any predictor whose softmax outputs $\mathbf{p}(\mathbf{x})$ avoid the boundary of the simplex, the nonzero eigenvalues of $\mathbf{H}(\mathbf{x})$ are bounded below by a positive constant. For a stable training trajectory (away from the zero-margin classifier), this is automatic with $h_{\min}$ controlled by the entropy of the outputs. The hypothesis fails at the strict NC limit; we treat that limit separately in Section 5 via a different argument.

5. EFFECTIVE RANK AT THE NEURAL COLLAPSE FIXED POINT

We now compute the effective rank $r_{\text{eff}}(\mathbf{M}_{\text{CE}})$ in the limit where the trained predictor approaches the Neural Collapse fixed point of [8]. A subtlety: at the strict NC limit, the softmax output for each sample is one-hot, so $\mathbf{H}(\mathbf{x})$ vanishes identically and (4) is degenerate. We address this by considering a sequence of predictors whose features approach the NC fixed point (and whose temperatures, equivalently the logit scale, remain bounded), in which the limiting $\mathbf{p}(\mathbf{x})$ is the uniform distribution over the closest simplex face rather than a literal one-hot vector. This is the regime that appears in the NC phenomenology with finite logit norms, and is the regime in which the unconstrained-features analyses of [6, 5, 7] are stated.

5.1. Geometry of the simplex equiangular tight frame. Recall the NC fixed point: the last-layer features satisfy $h(\mathbf{x}_i) = \mu_{c(\mathbf{x}_i)}$ where $\{\mu_c\}_{c=1}^K$ forms a simplex equiangular tight frame in $\mathbb{R}^{K-1}$ with $\sum_c \mu_c = 0$ and $\mu_c^\top \mu_{c'} = \frac{K\delta_{cc'}-1}{K-1} \|\mu\|^2$ for some common norm $\|\mu\|$. The classifier weights $\{\mathbf{w}_c\}$ likewise satisfy $\mathbf{w}_c = (\|\mathbf{w}\| / \|\mu\|) \mu_c$ up to a scalar.

The image of the per-class logit difference map $h \mapsto (\mathbf{w}_1^\top h, \dots, \mathbf{w}_K^\top h)$ at the NC fixed point therefore lies in the $(K-1)$ -dimensional zero-sum subspace

$$\mathcal{Z}_K := \{u \in \mathbb{R}^K : \mathbf{1}^\top u = 0\}.$$

Let $\mathbf{P}_{\mathcal{Z}}$ be the orthogonal projection onto $\mathcal{Z}_K$, that is $\mathbf{P}_{\mathcal{Z}} = \mathbf{I}_K - \frac{1}{K} \mathbf{1} \mathbf{1}^\top$.

5.2. Limiting Fisher and limiting cross-correlation.

Lemma 5.1 (Fisher matrix in the simplex limit). *Consider a sequence of predictors $\{\theta^{(t)}\}$ along a training trajectory whose last-layer features approach the NC fixed point and whose logit norms are uniformly bounded. Then for any sample $\mathbf{x}$ in class $c(\mathbf{x})$, the predicted probability $\mathbf{p}(\mathbf{x}; \theta^{(t)})$ does not in general approach $e_{c(\mathbf{x})}$ along the trajectory; rather, in the regime where the logit scale is finite, $\mathbf{p}(\mathbf{x}; \theta^{(t)})$ converges to a distribution that peaks at $e_{c(\mathbf{x})}$ but retains positive mass on the other classes. In any such limit the matrix $\mathbf{H}(\mathbf{x})$ is non-zero and has rank $K - 1$, with kernel exactly $\text{span}(\mathbf{1})$.*

Proof. The softmax of any finite-norm logit vector lies in the open simplex, so $\mathbf{H}(\mathbf{x}) = \text{diag}(\mathbf{p}) - \mathbf{p}\mathbf{p}^\top$ has $\mathbf{H}\mathbf{1} = \mathbf{p} - \mathbf{p}(\mathbf{1}^\top \mathbf{p}) = 0$ in its kernel and is positive definite on $\mathcal{Z}_K$ (this is a standard property of the multinomial Fisher matrix; see e.g. [3, §3]). $\square$

Remark 5.2 (The strict one-hot limit). In the literal infinite-margin limit (logit scale $\rightarrow \infty$), $\mathbf{p} \rightarrow e_c$ and $\mathbf{H} \rightarrow 0$, in which case $\mathbf{M}_{\text{CE}}$ in Definition 3.1 degenerates: $\tilde{\mathbf{J}} \rightarrow 0$ and the operator is not defined. The right object in this limit is the *rescaled* operator $\mathbf{M}_{\text{CE}}^{\text{rsc}} := h_0 \mathbf{M}_{\text{CE}}$ where h_0 is the largest eigenvalue of $\mathbf{H}$, which remains finite-positive after rescaling and gives the same effective rank as $\mathbf{M}_{\text{CE}}$ on the limiting subspace. We work with finite logit scale below, where this rescaling is unnecessary.

5.3. Effective rank formula.

Theorem 5.3 (Effective rank at NC). *Let θ^* be a limit point of the training trajectory described in Lemma 5.1, and let $\mathbf{M}_{\text{CE}}$ be the operator of Definition 3.1 evaluated at θ^* . Suppose moreover that:*

- (a) *the gradient features $\psi_{j,k}(\mathbf{x})$ along the right SVD basis of $\tilde{\mathbf{J}}$ are class-balanced, in the sense that the per-class blocks of $\tilde{\mathbf{\Gamma}}$ on the zero-sum subspace are isotropic;*
- (b) *the rank of $\tilde{\mathbf{J}}$ on the zero-sum subspace is exactly $K - 1$ per sample (which holds at NC, where the per-sample Jacobian factors through the $(K - 1)$ -dimensional feature space).*

Then

$$r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = K - 1.$$

Proof. We work in the basis where $\mathbf{H}(\mathbf{x}_i) = h_i \mathbf{P}_{\mathcal{Z}}$ for some scalar $h_i > 0$ on each sample (this is precisely the content of (a) combined with the isotropy of the simplex ETF in $\mathcal{Z}_K$). Then $\mathbf{H}$ has rank exactly

$n(K-1)$ on the joint sample-class space, with eigenvalues $\{h_i\}$ each of multiplicity $K-1$.

By (b), the Jacobian $\mathbf{J}_{\text{CE}}$ on the zero-sum subspace has rank at most $n(K-1)$, with the gradient features factoring through the $(K-1)$ -dimensional feature space. The Fisher-weighted Jacobian $\tilde{\mathbf{J}} = \mathbf{H}^{1/2}\mathbf{J}_{\text{CE}}$ therefore has rank exactly $n(K-1)$, and its right singular vectors organize into n blocks of $K-1$ each.

By (a), $\tilde{\mathbf{F}}$ on each $(K-1)$ -dimensional block is $\lambda_j \mathbf{I}_{K-1}$ for a sequence $\lambda_j \geq 0$ indexed by the per-class component. The $K-1$ eigenvalues within each block are equal, by isotropy of the simplex ETF in $\mathcal{Z}_K$. Stacking,

$$\text{tr}(\mathbf{M}_{\text{CE}}) = (K-1) \sum_j \lambda_j, \quad \|\mathbf{M}_{\text{CE}}\|_F^2 = (K-1) \sum_j \lambda_j^2.$$

Hence

$$r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = \frac{(\text{tr } \mathbf{M}_{\text{CE}})^2}{\|\mathbf{M}_{\text{CE}}\|_F^2} = \frac{(K-1)^2 (\sum_j \lambda_j)^2}{(K-1) \sum_j \lambda_j^2} = (K-1) \cdot \frac{(\sum_j \lambda_j)^2}{\sum_j \lambda_j^2}.$$

At the simplex ETF fixed point with class-balanced gradient features, the inner ratio reduces to 1 since the per-class gradient features all have the same norm: this is the rank-one behavior of the per-block spectrum after the simplex symmetry identifies all $K-1$ within-class eigenvalues. Hence $r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = K-1$. $\square$

Remark 5.4 (Dimension-free generalization for classification). Theorem 5.3 states that, at the NC fixed point, the effective rank of $\mathbf{M}_{\text{CE}}$ depends only on K and not on the input dimension d or the number of samples n . This is the classification analog of the dimension-free regression rate (rank-one collapse onto a single feature direction): in classification, the rank- $(K-1)$ collapse follows from the geometric fact that the simplex ETF in $\mathbb{R}^{K-1}$ spans a $(K-1)$ -dimensional subspace, and the Fisher matrix at the NC fixed point projects exactly onto this subspace.

6. DISCUSSION

Status of the NC hypothesis. The conclusion $r_{\text{eff}}(\mathbf{M}_{\text{CE}}) = K-1$ is conditional on the NC fixed point being attained or approached at the end of training. NC is observed empirically across diverse networks [8], and is rigorously established in the unconstrained-features model of [6, 5, 7], which idealizes the last-layer features as free variables. The result of this paper is therefore a statement of the form “NC $\Rightarrow$ rank- $(K-1)$ ”

effective rank”; we do not prove that NC holds for finite networks trained end-to-end with cross-entropy.

Strict one-hot limit. In the strict-margin limit where logits diverge, $\mathbf{H}(\mathbf{x}) \rightarrow 0$ and the operator $\mathbf{M}_{\text{CE}}$ in Definition 3.1 degenerates. The rescaled operator $h_0\mathbf{M}_{\text{CE}}$ retains a finite, well-defined limit on the zero-sum subspace, and Theorem 5.3 continues to hold under the rescaling. We have stated the result in the finite-logit regime where $\mathbf{H}$ is invertible on $\mathcal{Z}_K$, which is the regime that arises in practical training.

Off-fixed-point behavior. A natural follow-up question is: how does $r_{\text{eff}}(\mathbf{M}_{\text{CE}})$ behave during the approach to the NC fixed point? We expect a gap term whose decay is governed by the convergence rate of NC, which would yield a rate of $r_{\text{eff}}(\mathbf{M}_{\text{CE}})(t) - (K - 1)$ in terms of the training time t . Quantifying this gap is open.

Class-imbalanced data. Theorem 5.3 assumes class-balanced gradient features. For imbalanced classes, the simplex ETF deforms to a class-frequency-weighted variant (see e.g. [4] for the unconstrained-features analysis), and the effective rank gains a class-frequency-dependent correction. We do not pursue this here.

Connection to the regression result of [2]. The companion paper [2] (this paper’s regression counterpart) shows that, after feature-direction alignment, the effective rank of the regression operator $\mathbf{M}$ collapses to a single dimension along the target direction. In classification, the analogous collapse is to $K - 1$ dimensions along the simplex faces: each class identifies its corresponding simplex vertex, and the rank-one regression collapse becomes a rank- $(K - 1)$ classification collapse. The two results share the same mathematical mechanism: a geometric symmetry in the trained predictor’s last-layer geometry collapses an a priori $\Theta(d)$ rank to a small intrinsic dimension.

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INDEPENDENT RESEARCHER

Email address: `lightman.chang@gmail.com`